Started on	Sunday, 9 August 2026, 7:49 PM
State	Finished
Completed on	Sunday, 9 August 2026, 8:11 PM
Time taken	21 mins 33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

Reset answer

```
#include<stdio.h>

void func(int n)
{
    int count = 0;
    count++;
    if(n==1)
    {
        count++;
    }
    else
    {
        for(int i=1; count++, i<=n; count++, i++)
        {
            for(int j=1; count++, j<=n; j++)
            {
                count++;

                count++;

                break;
            }
        }
    }
    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);
    func(n);
```

```
33      return 0;
34  }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Sunday, 9 August 2026, 8:13 PM
State	Finished
Completed on	Sunday, 9 August 2026, 8:32 PM
Time taken	18 mins 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```c
#include<stdio.h>

void factor(int num)
{
    int count=0;
    int i=1;
    count++;

    for(i=1; i<=num; i++)
    {
        count++;
        count++;
        if(num%i == 0)
        {
            count++;
        }
    }
    printf("%d", count);
}

int main()
{
    int num;
    scanf("%d", &num);
    factor(num);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Wednesday, 26 August 2026, 9:48 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 9:59 AM
Time taken	11 mins 12 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```
#include<stdio.h>
void function(int n)
{
    int c=0;
    c++;
    for(int i=n/2; i<n; i++)
    {
        c++;
        for(int j=1; j<n; j=2*j)
        {
            c++;
            for(int k=1; k<n; k=k*2)
            {
                c++;
                c++;
            }
            c++;
        }
        c++;
    }
    c++;
    printf("%d\n", c);
}

int main()
{
    int n;
    if(scanf("%d", &n)==1 && n>0)
    {
        function(n);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔

	Input	Expected	Got	
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Wednesday, 26 August 2026, 10:24 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 10:27 AM
Time taken	3 mins 20 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```
1  #include<stdio.h>
2  int main()
3  {
4      int n, c=0;
5      int rev=0, remainder;
6
7      scanf("%d", &n);
8
9      while(n != 0)
10     {
11         c++;
12         remainder = n%10;
13
14         c++;
15         rev = rev*10+remainder;
16
17         c++;
18         n/=10;
19
20         c++;
21     }
22     c++;
23     c++;
24     c++;
25     printf("%d", c);
26     return 0;
27 }
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.